

# David Tong Standard Model Solutions — Sheet 1: The Basics

*D. Tong, The Standard Model, Example Sheet 1; 7 worked problems.*

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Quantum Field Theory

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*These prompts paraphrase David Tong's The Standard Model, Example Sheet 1 (January 2026). Problem numbers, starred status, equations, and guidance follow the official sheet.*

## 1. Sheet 1, Problem 1: Lorentz Generators for Weyl Spinors

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Define

$$\sigma^{\mu\nu} = \frac{i}{4}(\sigma^\mu \bar{\sigma}^\nu - \sigma^\nu \bar{\sigma}^\mu). \quad (1.1)$$

Verify that these matrices obey

$$[\sigma^{\mu\nu}, \sigma^{\rho\sigma}] = i(\eta^{\nu\rho} \sigma^{\mu\sigma} - \eta^{\nu\sigma} \sigma^{\mu\rho} + \eta^{\mu\sigma} \sigma^{\nu\rho} - \eta^{\mu\rho} \sigma^{\nu\sigma}). \quad (1.2)$$

*Hint:* Rewrite  $\sigma^{\mu\nu} = \frac{i}{2}(\eta^{\mu\nu} - \sigma^\nu \bar{\sigma}^\mu)$  and establish  $\sigma^\mu \bar{\sigma}^\nu + \sigma^\nu \bar{\sigma}^\mu = 2\eta^{\mu\nu} \mathbf{1}_2$ .

**Solution.** Recall  $\{\sigma_i, \sigma_j\} = 2\delta_{ij}$ , which follows

$$\sigma^\mu \bar{\sigma}^\nu + \sigma^\nu \bar{\sigma}^\mu = 2\eta^{\mu\nu} \mathbf{1}_2. \quad (1.3)$$

Hence,

$$\sigma^{\mu\nu} = \frac{i}{2}(\eta^{\mu\nu} - \sigma^\nu \bar{\sigma}^\mu), \quad (1.4)$$

which gives

$$[\sigma^{\mu\nu}, \sigma^{\rho\sigma}] = -\frac{1}{4}(\sigma^\nu \bar{\sigma}^\mu \sigma^\sigma \bar{\sigma}^\rho - \sigma^\sigma \bar{\sigma}^\rho \sigma^\nu \bar{\sigma}^\mu). \quad (1.5)$$

Now the strategy is to permute the indices using (1.3) so that the 4  $\sigma$  term eventually cancels. Carrying out the calculation, we find

$$[\sigma^{\mu\nu}, \sigma^{\rho\sigma}] = -\frac{1}{4}(-2\eta^{\rho\nu} \sigma^\sigma \bar{\sigma}^\mu + 2\eta^{\mu\rho} \sigma^\sigma \bar{\sigma}^\nu - 2\eta^{\nu\sigma} \sigma^\mu \bar{\sigma}^\rho + 2\eta^{\mu\sigma} \sigma^\nu \bar{\sigma}^\rho). \quad (1.6)$$

Then, plugging (1.4) in, we find

$$[\sigma^{\mu\nu}, \sigma^{\rho\sigma}] = i(\eta^{\nu\rho} \sigma^{\mu\sigma} - \eta^{\nu\sigma} \sigma^{\mu\rho} + \eta^{\mu\sigma} \sigma^{\nu\rho} - \eta^{\mu\rho} \sigma^{\nu\sigma}). \quad (1.7)$$

## 2. Sheet 1, Problem 2: Raised Weyl Indices and Lorentz Scalars

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Let  $S \in SL(2, \mathbb{C})$  act on left- and right-handed spinors by

$$(\psi_L)_\alpha \mapsto S_\alpha{}^\beta (\psi_L)_\beta, \quad (\psi_R)_{\dot{\alpha}} \mapsto (S^*)_{\dot{\alpha}}{}^{\dot{\beta}} (\psi_R)_{\dot{\beta}}. \quad (2.1)$$

The dotted index distinguishes the two representations. Prove the following statements in the sheet's epsilon convention:

- i)  $(S^{-1})_\beta^\alpha = \epsilon^{\alpha\gamma} S_\gamma^\lambda \epsilon_{\lambda\beta}$ .
- ii)  $(\psi_L)^\alpha = \epsilon^{\alpha\beta} (\psi_L)_\beta$  transforms with  $(S^{-1})_\beta^\alpha$  on its right.
- iii)  $\psi_L \chi_L = (\psi_L)^\alpha (\chi_L)_\alpha$  is Lorentz invariant.
- iv)  $(\psi_R)^{\dot{\alpha}} = \epsilon^{\dot{\alpha}\dot{\beta}} (\psi_R)_{\dot{\beta}}$  transforms with  $S^{-1\dagger}$ .
- v)  $\bar{\psi}_R \psi_L = (\psi_R^*)^\alpha (\psi_L)_\alpha$  is also a Lorentz scalar.

**Solution.**

i)

$$S_\alpha^\gamma S_\beta^\delta \epsilon_{\gamma\delta} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} 0 & ad - bc \\ -ad + bc & 0 \end{pmatrix} = \det S \epsilon_{\alpha\beta}. \quad (2.2)$$

Since  $S \in SL(2, \mathbb{C})$ ,  $\det S = 1$ , we find

$$S_\alpha^\gamma S_\beta^\delta \epsilon_{\gamma\delta} = \epsilon_{\alpha\beta} \implies \epsilon^{\beta\lambda} S_\alpha^\gamma S_\lambda^\delta \epsilon_{\gamma\delta} = \delta_\alpha^\beta \implies (S^{-1})_\beta^\alpha = \epsilon^{\alpha\gamma} S_\gamma^\lambda \epsilon_{\lambda\beta}. \quad (2.3)$$

ii)

$$(\psi_L)^\alpha = \epsilon^{\alpha\beta} (\psi_L)_\beta \rightarrow \epsilon^{\alpha\beta} S_\beta^\gamma (\psi_L)_\gamma = \epsilon^{\alpha\beta} S_\beta^\gamma \epsilon_{\lambda\gamma} (\psi_L)^\lambda = (S^{-1})_\gamma^\alpha (\psi_L)^\gamma. \quad (2.4)$$

iii)

$$\psi_L \chi_L \rightarrow (\psi_L)^\gamma (S^{-1})_\gamma^\alpha S_\alpha^\beta (\chi_L)_\beta = \psi_L \chi_L. \quad (2.5)$$

iv) Completely analogous to the left-handed case, we find

$$(\psi_R)^{\dot{\alpha}} \rightarrow (S^{-1\dagger})^{\dot{\alpha}}_{\dot{\beta}} (\psi_R)^{\dot{\beta}}. \quad (2.6)$$

v)

$$\bar{\psi}_R \psi_L \rightarrow (\psi_R^*)^{\dot{\alpha}} (S^{-1})_{\dot{\alpha}}^\alpha S_\alpha^\beta (\psi_L)_\beta = \bar{\psi}_R \psi_L. \quad (2.7)$$

**3. Sheet 1, Problem 3: The Right-Handed Lorentz Representation**

In addition to  $\sigma^{\mu\nu}$  above, define

$$\bar{\sigma}^{\mu\nu} = \frac{i}{4} (\bar{\sigma}^\mu \sigma^\nu - \bar{\sigma}^\nu \sigma^\mu). \quad (3.1)$$

Show  $\bar{\sigma}^{\mu\nu} = (\sigma^{\mu\nu})^\dagger$ . If

$$S = \exp\left(-\frac{i}{2} \omega_{\mu\nu} \sigma^{\mu\nu}\right), \quad (3.2)$$

show that

$$S^{-1\dagger} = \exp\left(-\frac{i}{2} \omega_{\mu\nu} \bar{\sigma}^{\mu\nu}\right). \quad (3.3)$$

It is enough to verify the infinitesimal transformation and then use the group law. Relate this to the dotted and undotted components of a Dirac spinor.

**Solution.**

$$(\sigma^{\mu\nu})^\dagger = -\frac{i}{4} (\bar{\sigma}^\nu \sigma^\mu - \bar{\sigma}^\mu \sigma^\nu) = \bar{\sigma}^{\mu\nu}. \quad (3.4)$$

For an infinitesimal transformation, we have

$$S^{-1\dagger} = 1 + \left(\frac{i}{2} \omega_{\mu\nu} \sigma^{\mu\nu}\right)^\dagger = 1 - \frac{i}{2} \omega_{\mu\nu} \bar{\sigma}^{\mu\nu}. \quad (3.5)$$

Thereby, the exponential map is given by

$$S^{-1\dagger} = \exp\left(-\frac{i}{2}\omega_{\mu\nu}\bar{\sigma}^{\mu\nu}\right). \quad (3.6)$$

#### 4. Sheet 1, Problem 4: Dirac and Majorana Mass Mixing

Two Weyl spinors have a real Dirac mass  $M$  and complex Majorana masses  $m_1, m_2$ :

$$\mathcal{L}_{\text{mass}} = -M(\bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L) + \frac{m_1}{2}\psi_L\psi_L + \frac{m_1^*}{2}\bar{\psi}_L\bar{\psi}_L + \frac{m_2}{2}\psi_R\psi_R + \frac{m_2^*}{2}\bar{\psi}_R\bar{\psi}_R. \quad (4.1)$$

Determine the physical fermion masses.

**Solution.** The mass Lagrangian can be written as

$$\mathcal{L}_{\text{mass}} = -\frac{1}{2}(\bar{\psi}_L \quad \psi_R) \begin{pmatrix} -m_1^* & M \\ M & -m_2^* \end{pmatrix} \begin{pmatrix} \bar{\psi}_L \\ \psi_R \end{pmatrix} + h.c. \equiv -\frac{1}{2}\chi_i \mathcal{M}_{ij} \chi_j + h.c., \quad (4.2)$$

where  $\chi^T \equiv (\bar{\psi}_L, \psi_R)$ . The kinetic part of the Lagrangian can be written as

$$\bar{\psi}_L i\bar{\sigma}^\mu \partial_\mu + \bar{\psi}_R i\sigma^\mu \partial_\mu \psi_R = \frac{1}{2}(\bar{\chi}_i i\bar{\sigma}^\mu \partial_\mu \chi_i + \chi_i i\sigma^\mu \partial_\mu \bar{\chi}_i). \quad (4.3)$$

The equation of motion is given by

$$i\sigma^\mu \partial_\mu \chi_i - \mathcal{M}_{ij}^* \bar{\chi}_j = 0 \quad (4.4)$$

and its conjugate

$$i\bar{\sigma}^\mu \partial_\mu \bar{\chi}_i - \mathcal{M}_{ij} \chi_j = 0. \quad (4.5)$$

Applying  $i\bar{\sigma}^\nu \partial_\nu$  to the first equation, we find

$$-\bar{\sigma}^\nu \sigma^\mu \partial_\mu \partial_\nu \chi_i - \mathcal{M}_{ij}^* i\bar{\sigma}^\nu \partial_\nu \chi_j = -\bar{\sigma}^{(\nu} \sigma^{\mu)} \partial_\mu \partial_\nu \chi_i - \mathcal{M}_{ij}^* \mathcal{M}_{jk} \chi_k = 0 \implies (\partial^2 + \mathcal{M}^\dagger \mathcal{M})\chi = 0. \quad (4.6)$$

To progress, let us diagonalise

$$\mathcal{M}^\dagger \mathcal{M} = U D U^\dagger. \quad (4.7)$$

Then, the equation of motion becomes

$$(\partial^2 + U D U^\dagger)\chi = 0 \implies (\partial^2 + D)(U^\dagger \chi) = 0. \quad (4.8)$$

Therefore, the physical masses squared are the eigenvalues of  $\mathcal{M}^\dagger \mathcal{M}$ . The mass eigenvalues are found to be

$$m_{\pm}^2 = \frac{1}{2} \left[ |m_1|^2 + |m_2|^2 + 2M^2 \pm \sqrt{(|m_1|^2 + |m_2|^2 + 2M^2)^2 - 4|m_1 m_2 - M^2|^2} \right]. \quad (4.9)$$

#### 5. Sheet 1, Problem 6\*: Discrete Symmetries of Gauge Fields

Use the chiral-basis matrices

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix}, \quad \gamma^5 = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & -\mathbf{1} \end{pmatrix}, \quad \sigma^\mu = (\mathbf{1}, \sigma^i), \quad \bar{\sigma}^\mu = (\mathbf{1}, -\sigma^i). \quad (5.1)$$

For a Dirac fermion coupled to  $U(1)$ , the gauge-field equation is  $\partial_\nu F^{\mu\nu} = e\bar{\psi}\gamma^\mu\psi$ . Starting from the spinor actions of  $C, P, T$ , work out how  $A_\mu$  must transform so that this equation remains invariant. Then determine how the Maxwell theta term

$$S_\theta = \int d^4x F_{\mu\nu} {}^* F^{\mu\nu} \quad (5.2)$$

transforms under each of  $C, P, T$ .

**Solution.** Under parity, we have

$$P : \quad \psi(t, \mathbf{x}) \rightarrow \gamma^0 \psi(t, -\mathbf{x}), \quad \bar{\psi} \gamma^\mu \psi(t, \mathbf{x}) \rightarrow \bar{\psi} \gamma^\mu \psi(t, -\mathbf{x}), \quad (5.3)$$

which implies

$$P : \quad \partial_\mu F^{\mu 0}(t, \mathbf{x}) \rightarrow \partial_\mu F^{\mu 0}(t, -\mathbf{x}), \quad \partial_\mu F^{\mu i}(t, \mathbf{x}) \rightarrow -\partial_\mu F^{\mu i}(t, -\mathbf{x}). \quad (5.4)$$

This follows

$$P : \quad \partial_i \partial^i A^0 - \partial_i \partial^0 A^i \rightarrow \partial_i \partial^i A'^0 + \partial_i \partial^0 A'^i \implies P : \quad A^0(t, \mathbf{x}) \rightarrow A^0(t, -\mathbf{x}), \quad A^i(t, \mathbf{x}) \rightarrow -A^i(t, -\mathbf{x}). \quad (5.5)$$

One can check that the spatial components also transform as desired.

Under charge conjugation, we have

$$C : \quad \psi(t, \mathbf{x}) \rightarrow \pm i \gamma^2 \psi^*(t, \mathbf{x}), \quad \bar{\psi} \gamma^\mu \psi(t, \mathbf{x}) \rightarrow -\psi^T \gamma^2 \gamma^0 \gamma^\mu \gamma^2 \psi^*(t, \mathbf{x}) = -\bar{\psi} \gamma^\mu \psi(t, \mathbf{x}), \quad (5.6)$$

where we have used  $\gamma^2 \gamma^\mu \gamma^2 = -\gamma^\mu$ . Therefore, we must have

$$C : \quad A^\mu(t, \mathbf{x}) \rightarrow -A^\mu(t, \mathbf{x}). \quad (5.7)$$

Under time reversal, we have

$$T : \quad \psi(t, \mathbf{x}) \rightarrow \gamma^1 \gamma^3 \psi^*(-t, \mathbf{x}), \quad \bar{\psi} \gamma^0 \psi(t, \mathbf{x}) \rightarrow \psi^T (\gamma^0)^* \psi^*(-t, \mathbf{x}) = \bar{\psi} \gamma^0 \psi(-t, \mathbf{x}), \\ \bar{\psi} \gamma^i \psi(t, \mathbf{x}) \rightarrow -\psi^T (\gamma^i)^* \psi^*(-t, \mathbf{x}) = -\bar{\psi} \gamma^i \psi(-t, \mathbf{x}). \quad (5.8)$$

The LHS of the gauge-field equation transforms as

$$T : \quad \partial_\mu F^{\mu 0}(t, \mathbf{x}) \rightarrow (\partial_j \partial^j A'^0 + \partial_j \partial^0 A'^j)(-t, \mathbf{x}), \quad \partial_\mu F^{\mu i}(t, \mathbf{x}) \rightarrow (\partial^2 A'^i + \partial_0 \partial^i A'^0 - \partial_i \partial^j A'^j)(-t, \mathbf{x}). \quad (5.9)$$

For the theory to be invariant under time reversal, we must have

$$T : \quad A^0(t, \mathbf{x}) \rightarrow A^0(-t, \mathbf{x}), \quad A^i(t, \mathbf{x}) \rightarrow -A^i(-t, \mathbf{x}). \quad (5.10)$$

The theta term is proportional to

$$\int d^4x \mathbf{E} \cdot \mathbf{B}, \quad (5.11)$$

whereas the fields transform as

$$P : \quad \mathbf{E}(t, \mathbf{x}) \rightarrow -\mathbf{E}(t, -\mathbf{x}), \quad \mathbf{B}(t, \mathbf{x}) \rightarrow \mathbf{B}(t, -\mathbf{x}) \quad (5.12)$$

$$C : \quad \mathbf{E}(t, \mathbf{x}) \rightarrow -\mathbf{E}(t, \mathbf{x}), \quad \mathbf{B}(t, \mathbf{x}) \rightarrow -\mathbf{B}(t, \mathbf{x}) \quad (5.13)$$

$$P : \quad \mathbf{E}(t, \mathbf{x}) \rightarrow \mathbf{E}(-t, \mathbf{x}), \quad \mathbf{B}(t, \mathbf{x}) \rightarrow -\mathbf{B}(-t, \mathbf{x}). \quad (5.14)$$

Therefore, the theta term is invariant under C, and odd under P and T.

## 6. Sheet 1, Problem 7: Parity with Scalar and Pseudoscalar Masses

Prove that  $\bar{\psi}\psi$  and  $i\bar{\psi}\gamma^5\psi$  are real. For

$$\mathcal{L}_{\text{mass}} = m_1 \bar{\psi}\psi + im_2 \bar{\psi}\gamma^5\psi, \quad (6.1)$$

rewrite the mass term in left- and right-handed Weyl fields using the chiral gamma matrices. Find an action of parity on  $\psi_L, \psi_R$  that preserves the theory, and express it on the Dirac spinor.

**Solution.**

$$(\bar{\psi}\psi)^* = \psi^T \gamma^0 \psi^* = \bar{\psi}\psi, \quad (6.2)$$

$$(i\bar{\psi}\gamma^5\psi)^* = -i\psi^T \gamma^0 \gamma^5 \psi^* = -i\psi^\dagger \gamma^5 \gamma^0 \psi = i\bar{\psi}\gamma^5\psi. \quad (6.3)$$

In Weyl spinors, the mass terms are

$$m_1(\psi_L^\dagger \psi_R + \psi_R^\dagger \psi_L) + im_2(\psi_L^\dagger \psi_R - \psi_R^\dagger \psi_L). \quad (6.4)$$

For the whole Lagrangian to be invariant under parity, we must have

$$P : \quad \psi_L(t, \mathbf{x}) \rightarrow e^{i\theta} \psi_R(t, -\mathbf{x}), \quad \psi_R(t, \mathbf{x}) \rightarrow e^{i\theta} \psi_L(t, -\mathbf{x}), \quad (6.5)$$

and hence

$$P : \quad \psi(t, \mathbf{x}) \rightarrow e^{i\theta} \gamma^0 \psi(t, -\mathbf{x}). \quad (6.6)$$

## 7. Sheet 1, Problem 8: Parity and Mass in 2 + 1 Dimensions

In signature  $(+, -, -)$  use imaginary matrices  $\gamma^\mu = (\sigma^2, i\sigma^1, i\sigma^3)$  and the action

$$S = - \int d^d x \left( i\bar{\psi} \gamma^\mu \partial_\mu \psi - M \bar{\psi} \psi \right). \quad (7.1)$$

Explain why parity reverses only  $x^1$  and leaves  $x^0, x^2$  unchanged, instead of sending the whole spatial vector to its negative. For  $M = 0$ , find parity, charge-conjugation, and time-reversal actions that preserve  $S$ . Which are broken by  $M \neq 0$ ?

**Solution.** We wish to use the parity action to enlarge  $SO(1, 2)$  to  $O(1, 2)$ . To do that, we need the determinant of parity to be negative. Therefore, this can only be achieved with one coordinate flipped in two spatial dimensions.

For parity, the massless Lagrangian transform as

$$P : \quad i\bar{\psi} \gamma^\mu \partial_\mu \psi \rightarrow i\psi^\dagger P^\dagger \gamma^0 (\gamma^0 \partial_0 - \gamma^1 \partial_1 + \gamma^2 \partial_2) P \psi. \quad (7.2)$$

Therefore, by choosing  $P : \psi \rightarrow \gamma^0 \psi$ , the massless theory remains invariant.

For charge conjugation, the Dirac field transforms as

$$C : \quad \psi \rightarrow C\psi^*, \quad (7.3)$$

while the kinetic term transforms as

$$C : \quad i\bar{\psi} \gamma^\mu \partial_\mu \psi \rightarrow i\psi^T C^\dagger \gamma^0 \gamma^\mu C \partial_\mu \psi^* = -i(\partial_\mu \psi^\dagger) C^T (\gamma^\mu)^T (\gamma^0)^T C^* \psi = i\psi^\dagger C^T \gamma^\mu \gamma^0 C^* \psi, \quad (7.4)$$

where we have used  $(\gamma^\mu)^T = -\gamma^\mu$ . Therefore, we must have  $C : \psi \rightarrow e^{i\alpha} \psi^*$ .

For time reversal, we have

$$T : \quad i \rightarrow -i, \quad \psi \rightarrow T\psi^*, \quad (7.5)$$

and hence

$$\begin{aligned} T : \quad i\bar{\psi} \gamma^\mu \partial_\mu \psi &\rightarrow -i\psi^T T^\dagger \gamma^0 (-\gamma^0 T \partial_0 - \gamma^1 T \partial_1 - \gamma^2 T \partial_2) \psi^* \\ &= -i\psi^\dagger T^T (-\gamma^0 \gamma^0 T^* \partial_0 - \gamma^1 \gamma^0 T^* \partial_1 - \gamma^2 \gamma^0 T^* \partial_2) \psi. \end{aligned} \quad (7.6)$$

For the kinetic term to be invariant, we must have  $T : \psi \rightarrow \gamma^1 \gamma^2 \psi^* = \gamma^0 \psi^*$ .

Now it is clear that the mass term is even under C, and odd under T and P.